

Project 05 – Secure EC2 Web Server

Objective

Deploy a secure Linux web server on Amazon EC2 using AWS Systems Manager for administration, Security Groups for network access, Apache HTTP Server for hosting, and a custom HTML/CSS portfolio website.

AWS Services Used

Service	Purpose
Amazon EC2	Virtual server hosting
Amazon VPC	Network environment
Security Groups	Firewall
IAM	Role permissions
Systems Manager	Secure administration
Amazon Linux 2023	Operating system
Apache HTTP Server	Web hosting

Key Activities

- Launch EC2 (Amazon Linux 2023).
- Attach IAM role for Systems Manager.
- Connect via Session Manager.
- Update Linux packages.
- Install and start Apache HTTP Server.
- Deploy custom HTML/CSS website.
- Configure HTTP and SSH Security Group rules.
- Verify public website using EC2 Public DNS.
- Troubleshoot networking and connectivity.

Skills Demonstrated

Amazon EC2 • IAM • AWS Systems Manager • Linux • Apache • Security Groups • Cloud Networking • Web Hosting • Troubleshooting

Outcome

Successfully deployed and secured a publicly accessible portfolio website on Amazon EC2, demonstrating practical cloud engineering, Linux administration, and AWS infrastructure management skills.